Lecture 24

Regular Sturm–Liouville problems

Objectives. *After a suitable positive shift, a regular second-order boundary problem has a symmetric Green kernel. Its inverse is compact on the correctly weighted Hilbert space. We prove the kernel construction and elementary eigenvalue properties. General completeness then follows from the cited compact spectral theorem.*

By the end of this lecture you will be able to:

- Retain the endpoint terms in Green's identity for Robin data.
- Prove reality, simplicity and weighted orthogonality of eigenfunctions.
- Construct the Green kernel with its signed flux jump.
- Deduce completeness with the stated compact-spectral qualification.

24.1 The regular weighted boundary problem

Fix a finite interval $[a,b]$. The coefficients are real, with $p \in C^1[a,b]$, $q, w \in C[a,b]$ and $p(x), w(x) > 0$ throughout. Define the differential expression and eigenvalue equation by

$$L[y] = -(p(x)y')' + q(x)y, \qquad L[y] = \lambda w(x) y.$$

In products and differential expressions we abbreviate $p = p(x)$, $q = q(x)$ and $w = w(x)$. In $L + c_0 w$, the weight denotes multiplication by $w(x)$. These abbreviations do not change the weighted measure. A classical solution has $y \in C^1[a,b]$ and $py' \in C^1[a,b]$. Since $p > 0$ is C^1, such a solution is also C^2. Impose real *separated Robin conditions*

$$\alpha_a y(a) + \beta_a p(a) y'(a) = 0, \qquad \alpha_b y(b) + \beta_b p(b) y'(b) = 0, \tag{24.1}$$

where each pair (α_a, β_a) and (α_b, β_b) is nonzero. Dirichlet means $y = 0$ at an endpoint. Neumann means $py' = 0$ there. These occur when the corresponding β or α is zero. An *eigenfunction* is a nonzero classical solution satisfying both conditions. Its parameter λ is an eigenvalue.

The weighted Hilbert space is $\mathcal{H} = L^2([a,b], w(x)\,\mathrm{d}x)$, with

$$\langle f, g \rangle_w = \int_a^b w(x) \overline{f(x)} g(x)\,\mathrm{d}x.$$

Write $\|f\|_w^2 = \langle f, f \rangle_w$ for its squared norm. The positive continuous weight has a positive minimum and finite maximum. Its norm is therefore equivalent to the usual L^2 norm, proving completeness from Theorem 5.3 and preserving smooth density. The weight is part of the problem and cannot be dropped from orthogonality.

The equation has a first-order system in the state (y, r_y), where $r_y(x) = p(x)y'(x)$ is the flux:

$$y' = r_y / p(x), \qquad r_y' = (q(x) - \lambda w(x)) y.$$

Its matrix is continuous on $[a, b]$. Extend that matrix constantly beyond each endpoint and translate any initial point to time zero. The continuous-coefficient linear-system result in § 2.6 then gives a unique solution on the whole interval, including endpoint initial data. For a complex parameter or complex initial state, split real and imaginary parts to obtain a real linear system of twice the dimension. Thus the same existence and uniqueness apply before we know that an eigenvalue is real.

24.2 Green's identity and eigenvalue properties

Theorem 24.1 (Green's identity). *For classical u and v,*

$$\int_a^b \left(\overline{u}L[v] - \overline{L[u]}v\right) \mathrm{d}x = \left[p(\overline{u}'v - \overline{u}v')\right]_a^b.$$

If both functions satisfy (24.1)*, the right-hand side is zero.*

Proof. Integrate each second-order term once by parts. Their interior derivative terms cancel and the real q terms cancel. The remaining endpoint bracket is the displayed expression. At each endpoint the pair (y, py') lies in the complex line spanned by the real vector $(\beta, -\alpha)$. For two such pairs, the skew product $\overline{py'_u}y_v - \overline{y_u}py'_v$ is zero. This proves the endpoint cancellation, including Dirichlet and Neumann cases. ∎

Proposition 24.2. *Every eigenvalue is real. Eigenfunctions for distinct eigenvalues are orthogonal in $\mathcal{H}$. Each eigenspace is one-dimensional.*

Proof. Apply Green's identity with the same eigenfunction in both places. It gives $(\lambda - \overline{\lambda})\,\|y\|_w^2 = 0$. Since $y \neq 0$ and $w > 0$, its weighted norm is positive, so λ is real. For eigenfunctions u and v with eigenvalues λ_m and λ_n, the same identity gives $(\lambda_n - \lambda_m)\,\langle u, v\rangle_w = 0$. Distinct parameters imply orthogonality.

At the left endpoint, allowed initial vectors $(y(a), p(a)y'(a))$ form a one-dimensional line. A solution with the zero initial vector is zero by linear ODE uniqueness. Two eigenfunctions with the same parameter have proportional initial vectors and hence are proportional everywhere. This uses the first-order construction given above, including its complex-state and endpoint initial-data cases. ∎

These boundary conditions cancel the skew bracket in Green's identity. They need not cancel the energy bracket $[\overline{y}py']_a^b$. That distinction matters when choosing an invertible real shift.

24.3 A positive shift

Define the quadratic expression on classical boundary-admissible functions by

$$Q[y] = \int_a^b \left(p|y'|^2 + q|y|^2\right) \mathrm{d}x - [\overline{y}py']_a^b.$$

Integration by parts gives $Q[y] = \int_a^b \overline{y}L[y]$. At a non-Dirichlet endpoint, the boundary condition gives $py' = -\alpha y/\beta$. Hence the endpoint contribution is a real multiple of $|y|^2$. At a Dirichlet endpoint it vanishes because $y = 0$. These real multiples can be negative.

For $0 < \delta \le b - a$ and $x \in [a, a+\delta]$, the fundamental theorem of calculus and Cauchy–Schwarz imply

$$|y(a)|^2 \le 2|y(x)|^2 + 2\delta \int_a^{a+\delta} |y'(t)|^2 \, \mathrm{d}t.$$

Average in x and enlarge the integrals to $[a, b]$ to obtain

$$|y(a)|^2 \le \frac{2}{\delta} \|y\|_2^2 + 2\delta \|y'\|_2^2 . \tag{24.2}$$

The same argument from the right gives the corresponding bound at b.

Lemma 24.3. *There is a real $c_0 > 0$ such that*

$$Q[y] + c_0 \|y\|_w^2 \ge \|y\|_2^2$$

for every classical function satisfying the boundary conditions. In particular, $(L + c_0 w)y = 0$ has no nonzero boundary-admissible solution.

Proof. Let $p_{\min} > 0$ and $w_{\min} > 0$ be the coefficient minima. The negative endpoint terms are bounded below by $-C(|y(a)|^2 + |y(b)|^2)$ for a fixed $C \ge 0$. Choose δ so small that $4C\delta \le p_{\min}/2$. The two trace bounds then give

$$Q[y] \ge \frac{p_{\min}}{2} \|y'\|_2^2 - C_1 \|y\|_2^2$$

for some $C_1 \ge 0$, including the bounded negative part of q. Choose $c_0 > 0$ with $c_0 w_{\min} - C_1 \ge 1$. This proves the estimate. For a homogeneous shifted solution, multiplying by $\overline{y}$ and integrating makes its left-hand side zero, forcing $y = 0$. ∎

The shift uses the full Robin data. Positivity of p and w alone would not justify discarding endpoint contributions.

24.4 The Green kernel

Choose real solutions u_l and u_r of $(L + c_0 w)u = 0$ with initial vectors $(\beta_a, -\alpha_a)$ at a and $(\beta_b, -\alpha_b)$ at b, respectively. Linear ODE theory supplies them on the whole finite interval. Each satisfies its own endpoint condition and is nonzero. The *Wronskian* of two functions is $W(u, v) = uv' - u'v$. Its arguments distinguish it from the domain of the flow W in Lecture 2. Differentiating and using their common equation shows that

$$W_0 = p(x) W(u_l, u_r)(x)$$

is constant. It is nonzero: if it vanished, the two solution vectors would be proportional at one point and hence everywhere. The resulting nonzero solution would satisfy both endpoints, contradicting Lemma 24.3. Define the Green kernel by

$$G(x, t) = -\frac{u_l(\min(x, t)) u_r(\max(x, t))}{W_0}. \tag{24.3}$$

The formula makes G continuous, real and symmetric. For fixed $t \in (a, b)$, it satisfies the endpoint conditions in x and the homogeneous equation away from $x = t$. Its flux jump is

$$p(t)\big[G_x(t+, t) - G_x(t-, t)\big] = -\frac{p(t)(u_l u_r' - u_l' u_r)(t)}{W_0} = -1. \tag{24.4}$$

This is the sign required by the leading minus in L.

Theorem 24.4. *For continuous f on $[a,b]$, the unique classical solution of $(L+c_0w)y = wf$ with the boundary conditions is*

$$(Tf)(x) = \int_a^b G(x,t)w(t)f(t)\,\mathrm{d}t.$$

Proof. Split the integral at $t = x$:

$$\begin{aligned}(Tf)(x) = -\frac{u_r(x)}{W_0}\int_a^x u_l(t)w(t)f(t)\,\mathrm{d}t \\ -\frac{u_l(x)}{W_0}\int_x^b u_r(t)w(t)f(t)\,\mathrm{d}t.\end{aligned}$$

Differentiate once. The endpoint terms cancel because both contain $u_l(x)u_r(x)w(x)f(x)$ with opposite signs. After multiplying that derivative by p and differentiating again, the new endpoint term is $-w(x)f(x)$ by the weighted Wronskian formula. The remaining terms satisfy the homogeneous shifted equation. Thus $(L + c_0w)Tf = wf$. The first-derivative formula also verifies both endpoint conditions, including its limiting values at a and b. All integrands are continuous, so these differentiations are justified. Uniqueness follows by subtracting two solutions and using the shift lemma. ∎

The kernel construction only required that the homogeneous shifted problem have no nonzero admissible solution. For Dirichlet data and $q(x) \geq 0$, this already holds without a shift: the identity

$$\int_a^b \overline{y}L[y]\,\mathrm{d}x = \int_a^b (p|y'|^2 + q|y|^2)\,\mathrm{d}x$$

is strictly positive for nonzero boundary-admissible y. Indeed, equality would give $y' = 0$, and the Dirichlet values would then give $y = 0$. Thus one may set $c_0 = 0$ in the same construction in this case. For $L = -\mathrm{d}^2/\mathrm{d}x^2$, choose $u_l(x) = x - a$ and $u_r(x) = b - x$. Their weighted Wronskian is $W_0 = -(b-a)$, giving the unshifted kernel

$$G_0(x,t) = \frac{(\min(x,t) - a)(b - \max(x,t))}{b-a}. \tag{24.5}$$

In particular, $G_0(1/3, 3/4) = 1/12$ on $[0,1]$.

The next diagram shows this kernel's matching and negative flux jump.

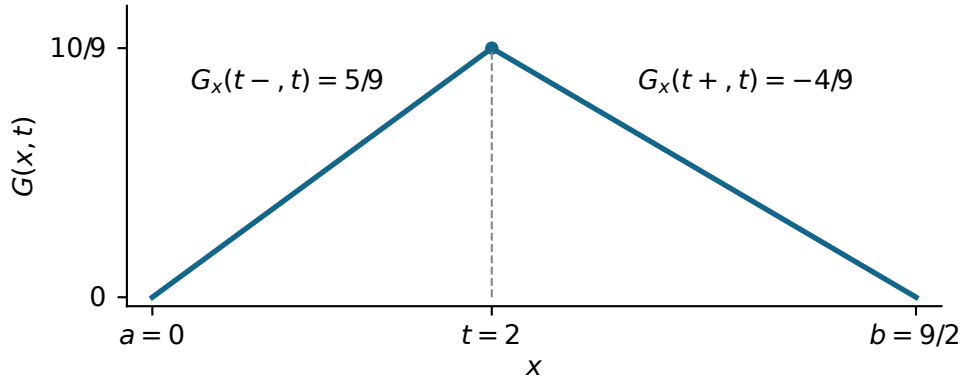


Figure 24.1: For the Dirichlet operator $-\mathrm{d}^2/\mathrm{d}x^2$ without a shift, the Green kernel is a tent; the drawing uses $a = 0$, $b = 9/2$ and $t = 2$. The general kernel has the same continuity and negative unit flux jump, but need not be piecewise linear.

The same construction can retain a non-Dirichlet endpoint explicitly. Solve the specified forced problem

$$-y'' = 2x \quad (0 \le x \le 1), \qquad y'(0) = 0, \qquad y'(1) + y(1) = 0.$$

A homogeneous solution is affine. The left condition makes it constant, and the right condition then makes it zero. Thus the construction is invertible without a shift. The endpoint solutions and their Wronskian are

$$u_l(x) = 1, \qquad u_r(x) = 2 - x, \qquad W_0 = -1.$$

Each solves the homogeneous equation and its own endpoint condition. The kernel formula gives

$$G(x,t) = 2 - \max(x,t).$$

For $x < t$ its derivative in x is zero. For $x > t$ that derivative is minus one. Thus the kernel is continuous, has the required jump, and satisfies both endpoint conditions. Applying it to the forcing gives

$$\begin{aligned} y(x) &= (2-x)\int_0^x 2t\,\mathrm{d}t + \int_x^1 (2-t)2t\,\mathrm{d}t \\ &= (2-x)x^2 + \left[2t^2 - \frac{2}{3}t^3\right]_x^1 = \frac{4}{3} - \frac{x^3}{3}. \end{aligned}$$

Applying the negative second derivative gives the prescribed forcing. At the left endpoint the derivative is zero. At the right endpoint the derivative is minus one and the value is one, so the Robin sum is zero. The integral formula therefore reconstructs the complete boundary solution.

24.5 Compactness and qualified completeness

The same kernel defines a bounded operator on $\mathcal{H}$. Indeed, Cauchy–Schwarz gives $|Tf(x)| \le \|G(x,\cdot)\|_w \, \|f\|_w$. Continuity of G on the compact square makes the first factor uniformly

bounded. It also makes Tf continuous, because $\|G(x,\cdot) - G(x_0,\cdot)\|_w \to 0$ as $x \to x_0$. Symmetry and Fubini give $\langle f, Th\rangle_w = \langle Tf, h\rangle_w$. Fubini applies since the bounded kernel multiplies two $L^1(w\,\mathrm{d}x)$ functions; finite measure and Cauchy–Schwarz give that integrability.

Uniform continuity approximates G uniformly by kernels constant on each rectangle of a finite partition. Their operators have finite-dimensional range, spanned by the interval indicators. A uniform kernel error ε gives operator-norm error at most $\varepsilon \int_a^b w$ by the same Cauchy–Schwarz estimate. Hence T is compact. To justify the last implication, take a bounded sequence and extract nested subsequences on which the images under each finite-rank approximant converge. On the diagonal subsequence, the operator-norm error makes the T-images Cauchy. Completeness supplies their limit, as required by the definition of compactness.

The null space of T consists only of the zero function. This is the set of inputs sent to zero, distinct from the integral kernel $G(x,t)$ used to define the operator. For $\phi \in C_c^\infty((a,b))$, put $h = (L + c_0 w)\phi/w$, which is continuous. The classical inverse theorem gives $Th = \phi$. If $Tf = 0$, self-adjointness gives $\langle f, \phi\rangle_w = \langle f, Th\rangle_w = \langle Tf, h\rangle_w = 0$. Smooth density, Theorem 5.9, and equivalence of the weighted norm force $f = 0$.

Theorem 24.5 (Regular spectral completeness · *sketched*)**.** *Under the stated regular coefficients and separated real boundary conditions, the eigenfunctions form an orthonormal basis in $\mathcal{H}$ after normalisation. The eigenvalues can be listed as $\lambda_1 < \lambda_2 < \cdots \to +\infty$. This conclusion uses the cited compact self-adjoint spectral theorem.*

Deduction from the cited compact theorem. Apply Theorem 18.8 to the compact self-adjoint injective operator T. Its orthonormal eigenbasis has nonzero eigenvalues μ_j. If $Tf = \mu f$, then $f = \mu^{-1} Tf$ has a continuous representative. Applying the classical inverse result to this continuous f proves that f is classical and satisfies

$$L[f] = (\mu^{-1} - c_0) w f.$$

The shift estimate gives $\mu^{-1}\|f\|_w^2 = Q[f] + c_0\|f\|_w^2 > 0$. Thus $\mu > 0$. Conversely, the shift estimate gives $\lambda + c_0 > 0$ for any classical eigenfunction, which therefore satisfies $Tf = (\lambda + c_0)^{-1} f$ by uniqueness of the shifted boundary problem. The two eigenfunction families therefore coincide. Moreover, if a T-eigenvalue μ differed from every basis eigenvalue μ_j, self-adjointness would make its eigenvector orthogonal to every basis vector. Density of their span would force that vector to be zero. Hence every classical eigenvalue occurs in the displayed list.

Each eigenspace is one-dimensional by the earlier ODE argument. The infinite-dimensional space $\mathcal{H}$ requires infinitely many basis vectors. Compactness allows only finitely many μ_j above any positive threshold. Order them decreasingly towards zero. Then $\lambda_j = \mu_j^{-1} - c_0$ increases strictly to infinity. Completeness is exactly the conclusion supplied by the cited input. ∎

24.6 The sine basis recovered

For $p = w = 1$ and $q = 0$, impose Dirichlet data on $[0, \ell]$, with $\ell > 0$. Here ℓ denotes the interval length, since L denotes the operator. Reality and the energy integral give $\lambda > 0$ for a nonzero eigenfunction. Solving $-y'' = \lambda y$ and imposing both zeros gives

$$\lambda_n = (n\pi/\ell)^2, \qquad u_n(x) = \sqrt{\frac{2}{\ell}}\sin(n\pi x/\ell), \qquad n \geq 1.$$

These are exactly the modes of Theorem 6.6. Their completeness was proved there independently, so this special case does not require the cited compact theorem. With constant weight $w = 2$ on $[0, 2]$, the equation instead gives $\lambda_1 = \pi^2/8$. The frequency in the sine and the weighted eigenvalue are distinct quantities.

Summary.

- Real separated Robin conditions cancel the skew boundary form. They give real simple eigenvalues and weighted orthogonality.
- Endpoint trace bounds permit a positive shift despite possibly negative Robin energy terms.
- Two homogeneous solutions give a continuous symmetric Green kernel with flux jump -1. Its weighted integral operator is compact and injective.
- General completeness inherits the cited compact spectral theorem. The Dirichlet sine case also has the independent proved Fourier argument.

Singular endpoints require additional hypotheses and are outside this result.

References

N. Jacob and K. P. Evans, *A Course in Analysis*, Vol. IV, 2018 printing, Chapter 21, pp. 355–378, and Vol. V, 2020 printing, Chapter 18, pp. 503–514. Our leading differential sign is negative; the Robin energy argument above retains its endpoint terms explicitly.